

Broadcast telecommunications

CMS.100

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Assessment guide (will be posted on website)

- 500 word guideline is a floor, not a ceiling
- **Dataset and gap**
- **Protocol in practice**
- **Data structure**
- **Instrument and process**
- **Claims**
- **Visualization**

Grading guidelines

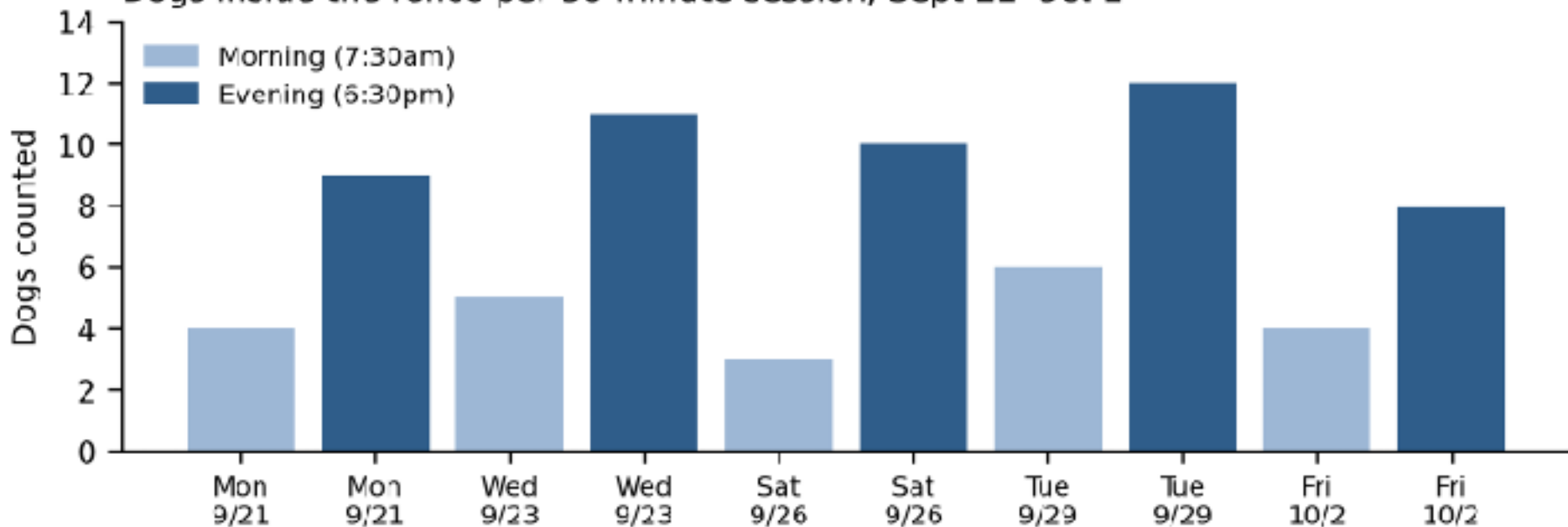
- **A:** each choice is deliberate and well-explained; argument is supported by the data and account of dataset is reflective and well-reasoned. The visualization makes an argument that the prose could not do alone.
- **B:** choices are sound and mostly explained but the reasoning is uneven. Does not account for how the measurement instrument shaped result or does not make a clear, complete argument
- **C:** makes arguments that the protocol / data don't support; incomplete or not fully explained or reflected upon
- **Below C:** doesn't follow the assignment instructions.

How did you rank them?

Exemplar	Ranked 1st	Ranked 2nd	Ranked 3rd
Dogs			
Grass			
Quotes			

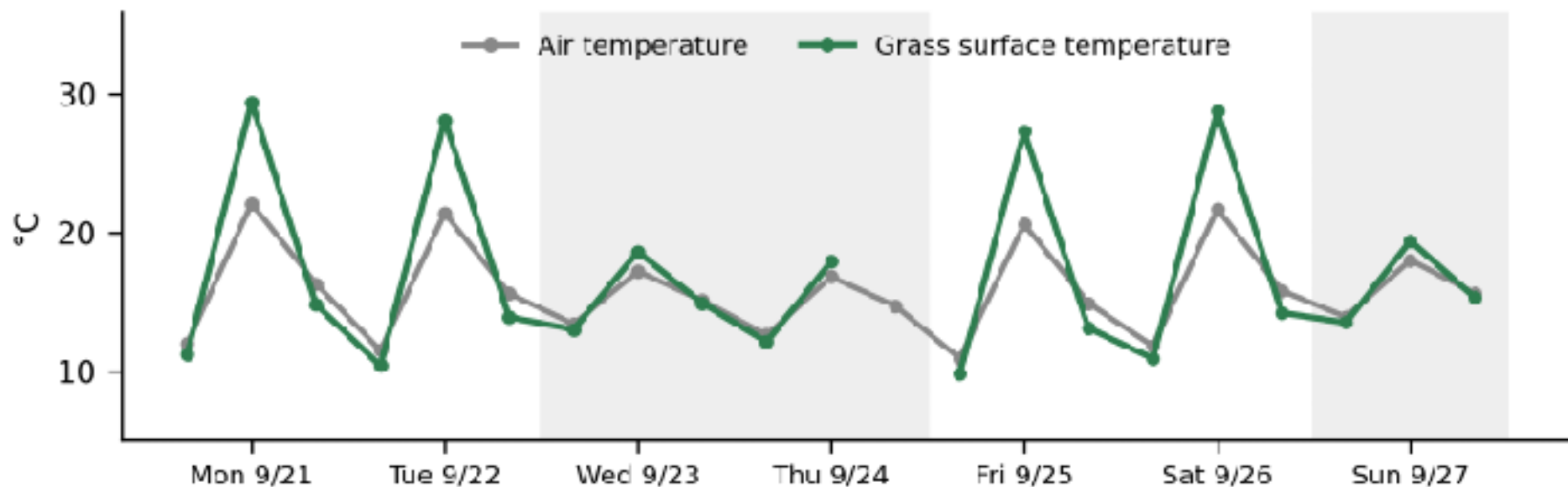
Dogs

Dogs inside the fence per 30-minute session, Sept 21-Oct 2

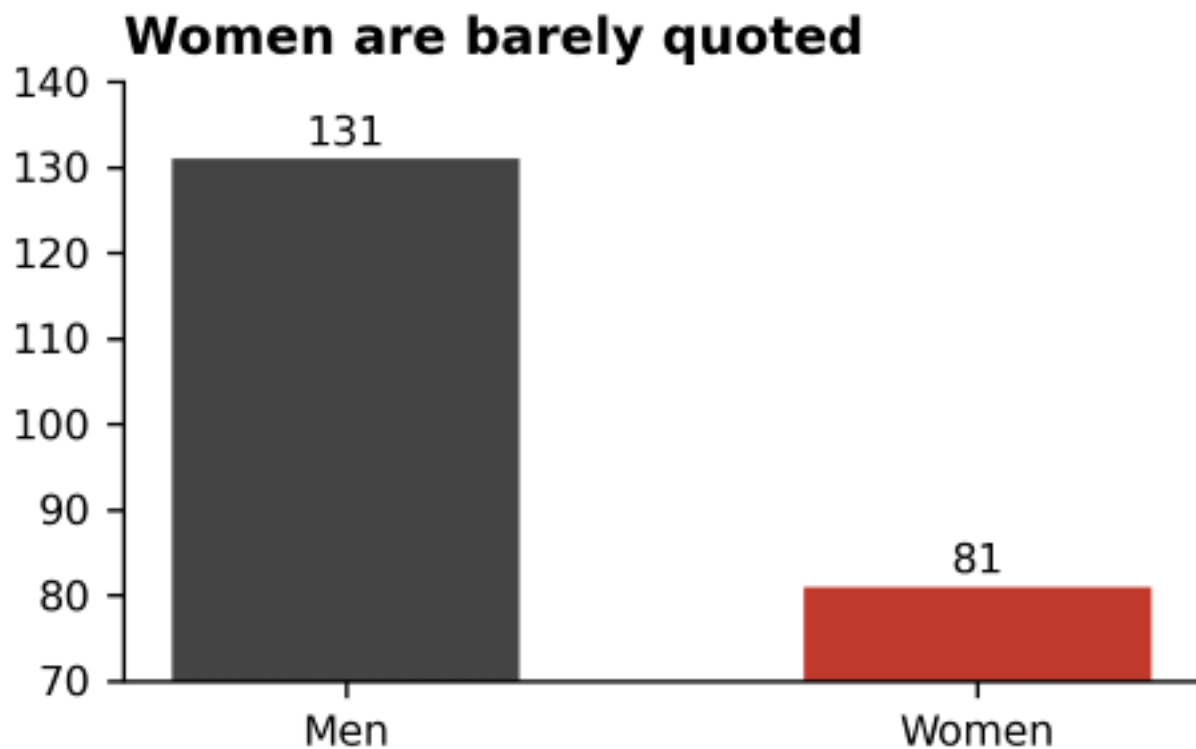


Grass

Grass surface vs. air temperature, three readings a day (shaded = cloudy)



Quotes



Logistics

- **A3.1:** presentation dates assigned Friday; pitches Oct 7 and 9
- **A1 due:** 9 am, Fri Oct 16
- Assessment guide posted online soon

CMS.340: Assessment Guide for A1

The purpose of this assignment is to understand how every dataset—regardless of field—is the result of decisions that someone made; it is not something that is simply “discovered.” In this case, that someone is you! In terms of assessment, I read your work as a whole and I use the following criteria to explain how successful (or not) the paper was in describing these different dimensions. There is not a checklist for which there are certain points attributed to each section. For C1–I purposes, the 500-word guideline is a floor, not a ceiling, so exceeding the word guideline is fine (but falling short is not).

Dataset and gap. Double’s project shows that the data we don’t collect says something about what we value. Your write-up should explain why the dataset matters—not just that it exists...and describe how and why your dataset addresses this gap.

Process in practice. Your protocol was a preliminary plan, and the actual collection process rarely goes exactly to plan—this is not only fine but expected! However, in the final write-up, a reader should be able to understand how you actually collected the data and where you found difficulties and what you did to address this.

Data structures. Someone else should be able to open your file and see it, which means that you must provide a guide to understanding how the dataset is structured (i.e., what it has, or “fields”; mean; what each variable represents.) You can do so in the form of a separate document (as with an appendix) or within the text of your write-up.

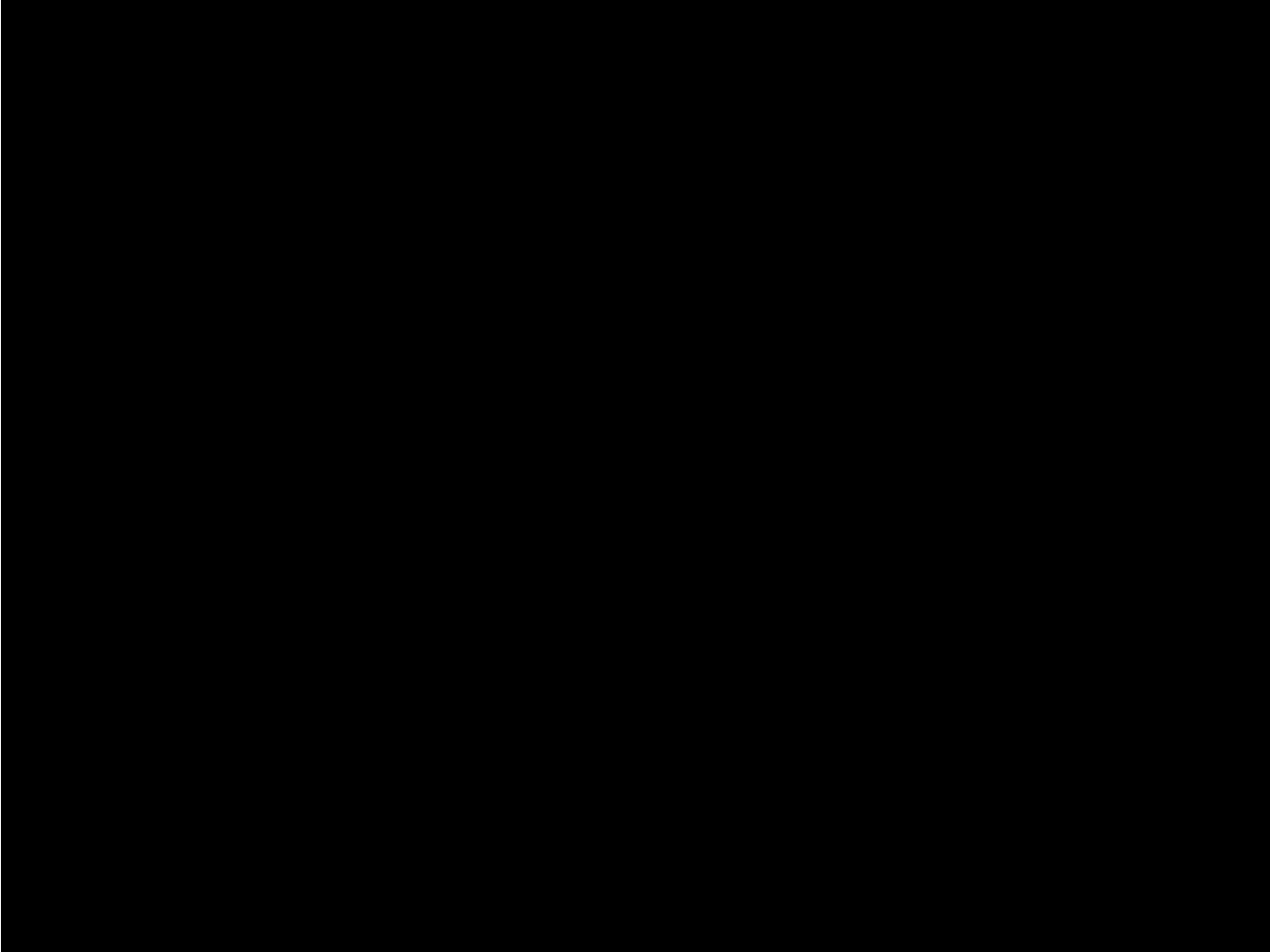
Instrument and process. Every measuring instrument shapes what it records—what did yours capture and what did it miss? What kinds of decisions did you have to make in the collection process? Were there uses, misuses, errors and how did you manage them?

Claims. A dataset supports some claims and not others. What does your dataset show and how does the dataset support the claim? What are its limitations? Place this in broader historical or social context.

Visualization. The visualization is labeled, constructed, and presented in a way that supports or otherwise corroborates the claims you make in the write-up. The visualization contributes to your broader argument; it is not just a statement of the main point.

How then, become a grade?

- **A:** Each choice in the dataset (e.g., what events are data, what is included) is deliberate and well-explained. The write-up presents an argument that is supported by the data and the account of how the dataset is constructed is reflexive and well-reasoned. The visualization makes an argument that the prose could not do alone.
- **B:** The choice is sound and mostly explained; the claims are generally supported by the data. However, the reasoning is uneven and the write-up does not fully account for how the measurement instrument shaped the results or does not make a clear, complex argument.
- **C:** The write-up mentions things the protocol could not have captured; construction of dataset is inconsistent; it is not fully explained or presented in the write-up.
- **Below C:** The dataset or write-up doesn’t follow the assignment instructions.



Reading check-in: pick one

On p. 179, Gitelman and Mullaney ask, “What is Morse Code without letters?” Why does this question matter?

Double mediation (p. 180): explain it, then apply it to something you use today.

What is infrastructure?

Embedded

sunk into other structures, social arrangements, and technologies

Transparent

invisibly supports tasks; you don't reinvent it each time

Built on an installed base

inherits the strengths and limits of what came before

Visible upon breakdown

you notice it when the server is down or the bridge washes out

“Optical fibers run along old railroad lines.”

What is semiotics?

The study of signs: how a form gets attached to a meaning.



Telegraphy was “in equal measure ... a linguistic and semiotic infrastructure.”

Infrastructure is also a sign

“Infrastructures are built networks that facilitate the flow of goods, people, or ideas and allow for their exchange over space.”

Brian Larkin, “The Politics and Poetics of Infrastructure,” *Annual Review of Anthropology* (2013), p. 328

Politics:

who builds it, who pays, who gets connected

Poetics:

what it promises (modernity, progress, speed) apart from what it does

Communication leaves transportation behind

Before

A message moves as fast as a horse, a train, or a ship

Speculation is about place: which market will pay more?

After

Associated Press (1846): news becomes a commodity sold by wire

Chicago Board of Trade (1848): speculation moves into time, as futures

“Messages ... could finally be sent without being physically carried.”

But the wires follow the old roads

1840s–1870s

Telegraph lines run along railroads and colonial trade routes

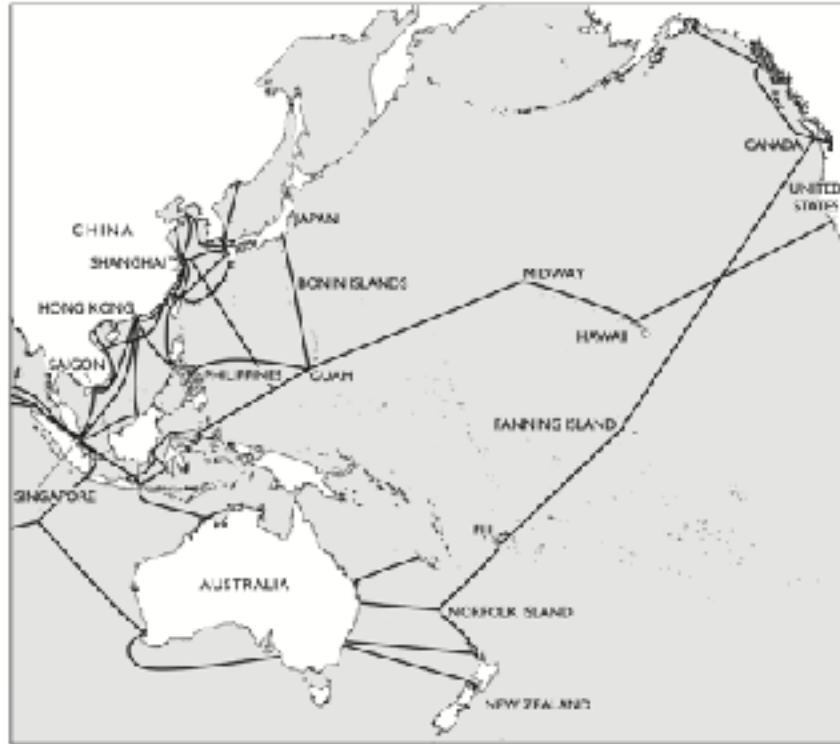
2010

Spread Networks digs an 825-mile fiber line, Chicago to New Jersey, to shave milliseconds off trades

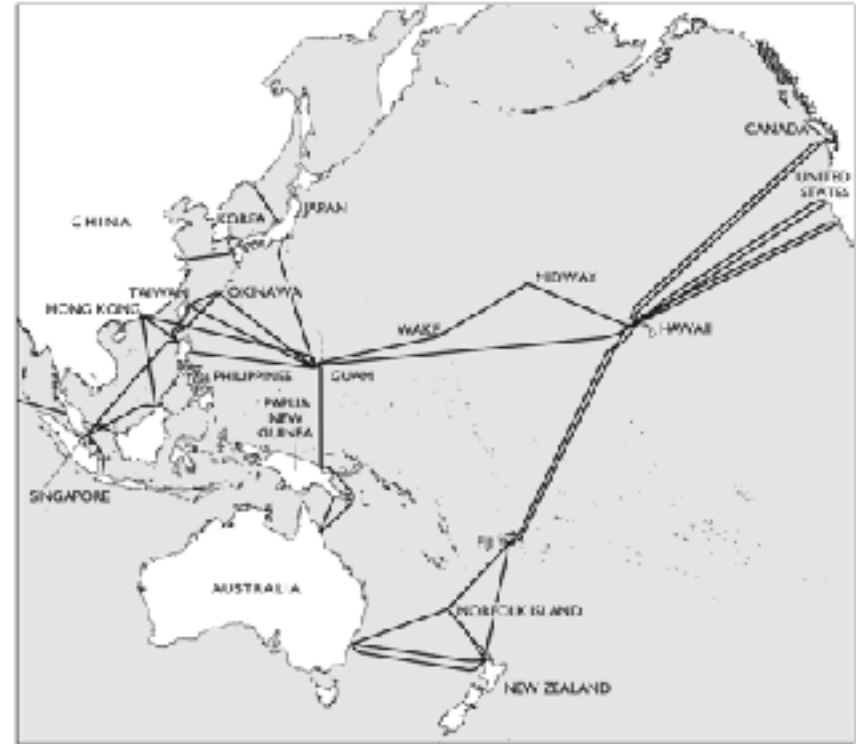
Today

“Fiber-optic cables for internet transmissions follow many of the same routes as the original telegraph lines” (G&M, p. 178)

How does the internet work? (A pre-history)



MAP 1.1. Transpacific cable routes, 1922.

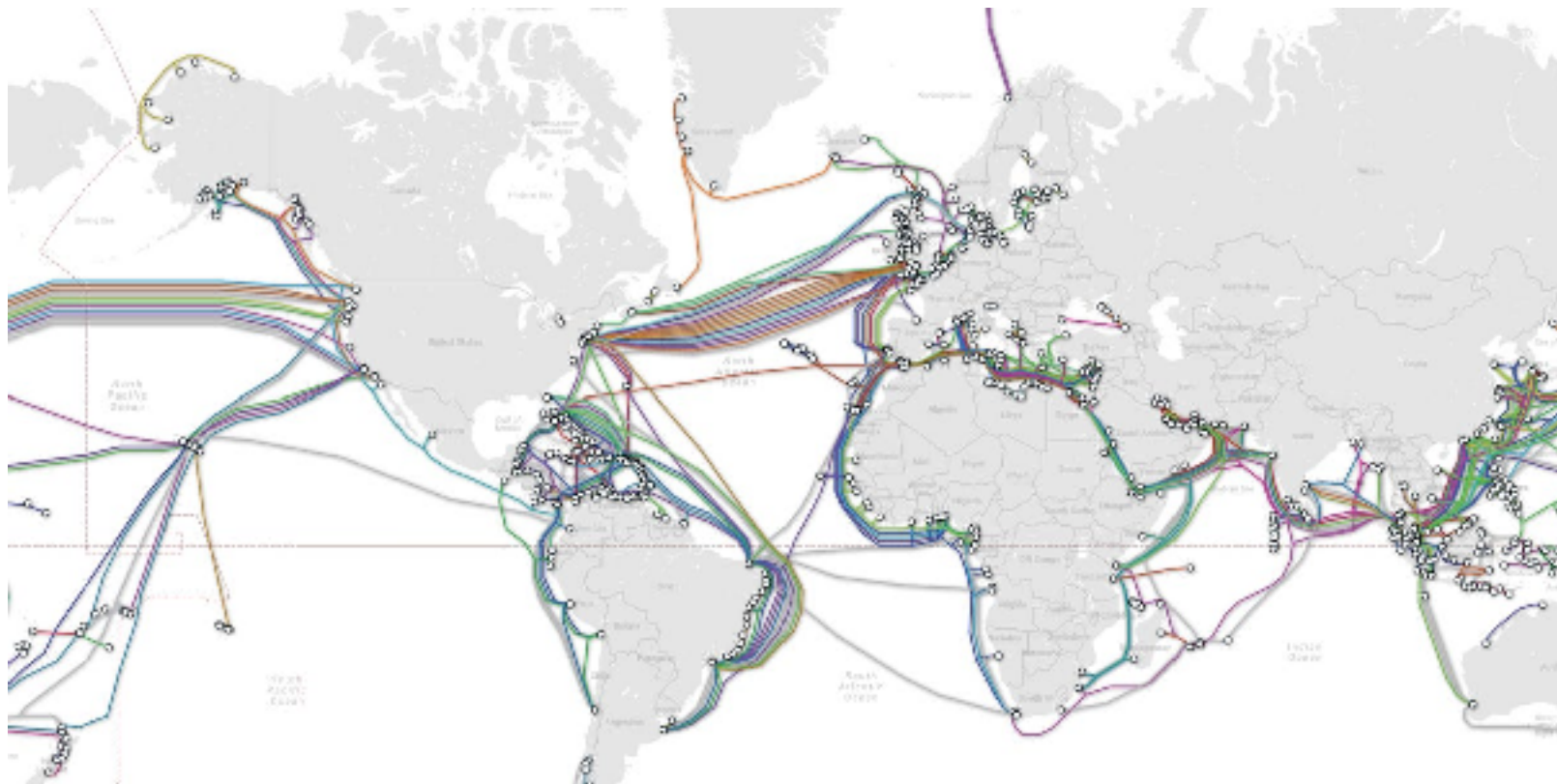


MAP 1.2. Transpacific cable routes, 1982.

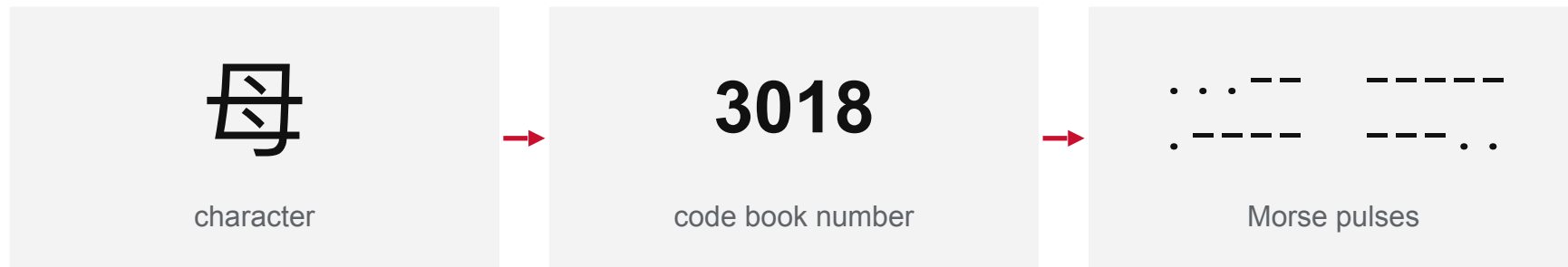
EASTERN TELEGRAPH C^d SYSTEM AND ITS GENERAL CONNECTIONS.



Undersea network cables



What is Morse code without letters?



1871: Schjellerup and Viguier's code numbers about 6,800 characters in Kangxi Dictionary order

Double mediation: character → Arabic numerals → Morse

Chinese became a “numbered language,” taxed like a cipher

Who is the code for?

Signs permitted by the International Telegraph Union, Vienna 1868:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

0 1 2 3 4 5 6 7 8 9

. , ; : ? ! ' + - / = () & « é

1875: accented é joins the alphabet; Morse users may send Ä Á Å Ñ Ö Ü

1903: those accented letters become part of the standard set

German and French overflowed the code. Chinese had no letters at all.

Translating telegram messages...

- Prices in Liverpool have fallen for three days running, and there is talk that the Russian harvest is much larger than expected. I think we should sell all the wheat we are holding before the rest of the market hears the news. Please do this at once and do not wait for my letter.
- Our ship reached New York this morning after a very rough crossing. Everyone is well, though Mother was seasick for most of the voyage and has not yet recovered her appetite. We will write a proper letter once we have found lodgings. Please tell Aunt Mary not to worry.
- Hélène est arrivée à Québec hier soir. Elle loge chez sa cousine, rue Saint-Jean, et vous écrira dimanche. (Hélène arrived in Québec last night. She is staying with her cousin on rue Saint-Jean and will write to you on Sunday.)

Telegrams cost 10¢ a word, and each message has a budget of 30¢, which is 3 words. A word longer than 15 letters counts as two, loosely following the 1875 international rules. Numbers count as words.

An “instant” medium ran on skilled labor

Twelve letters took you how long? How many came through wrong?

Operators decoded by ear, “on the fly,” all day (p. 176)

New industrial ailments: “telegrapher's palsy” (p. 177)

Strip C: where did È and À go? Strip D: how many steps before you could send anything?